Ventilation as a bilinear dynamical system

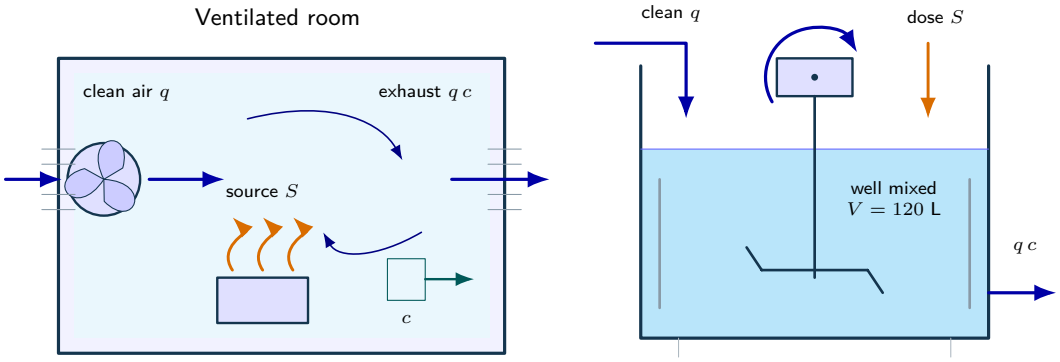
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

A hypothetical well-mixed room has $V = 120\,\mathrm{m}^3$, tracer generation $S = 360\,\mathrm{ppm\,m}^3/\mathrm{min}$ and nominal airflow $q_0 = 3\,\mathrm{m}^3/\mathrm{min}$. Inlet excess concentration is zero. Derive the equilibrium and local airflow-to-concentration model from $V\dot{c} = S - qc$. Compare 10% and 100% airflow increases using the nonlinear and linearized models, including their final-value errors.

2 Physical system



Inventory and channels
c: tracer concentration.
S: source rate; q: throughput.
Equal inlet/outlet flow fixes V.

Balance and bilinearity
$V\dot{c} = S - qc$.
The product qc couples state and input. Room and liquid cases use different volume units.

Mixing assumption
Grilles and baffles illustrate ideal mixing only. Air velocity fields, fan dynamics and mixing transients are not modeled.

A ventilated room and a well-mixed liquid analogue share an inventory balance. The room uses m^3 and the liquid case uses L; do not mix their concentration or flow units. Equal inlet and outlet volume flows keep each inventory constant.

3 Concentration balance

For a perfectly mixed room, let c be excess tracer concentration in ppm, q ventilation flow in $\mathrm{m}^3/\mathrm{min}$ and S tracer generation in ppm $\mathrm{m}^3/\mathrm{min}$. Conservation gives

$$V\dot{c} = S - qc.$$

The model is bilinear in the two variables q and c. At constant airflow $q_0 = 3$, the equilibrium is $c_0 = S/q_0 = 120$ ppm.

4 Local perturbation model

Write $c = c_0 + \delta c$ and $q = q_0 + \delta q$. Expanding the product and subtracting the equilibrium gives

$$V\delta\dot{c} = -q_0\delta c - c_0\delta q - \delta q\delta c.$$

Discarding only the second-order product yields

$$\frac{\delta C(s)}{\delta Q(s)} = \frac{-c_0/q_0}{(V/q_0)s+1} = \frac{-40}{40s+1}.$$

Increasing airflow reduces concentration, so the negative static gain has a physical meaning.

5 Finite changes and approximation error

For a flow step Δq, the exact response is

$$c(t) = \frac{S}{q_0+\Delta q} + \left(c_0 - \frac{S}{q_0+\Delta q}\right) e^{-(q_0+\Delta q)t/V}.$$

The local prediction is $c_{lin}(t) = c_0 - 40\Delta q(1-e^{-t/40})$. For a 10 percent step the exact final value is 120/1.1 ppm, while the linear prediction is 108 ppm. For a doubling, they are 60 ppm and 0 ppm: a useful warning against extrapolating the tangent model.

6 Interpretation

The linearized model changes neither its time constant nor its input gain after a step. The nonlinear model changes both its equilibrium and decay rate. This is a tracer-transport teaching example, not an indoor-air-quality exposure assessment.

7 What to learn from the experiment

The doubled-flow experiment deliberately leaves the small-signal regime. Its large error is a limitation of linearization, not a numerical failure.

8 System sketch

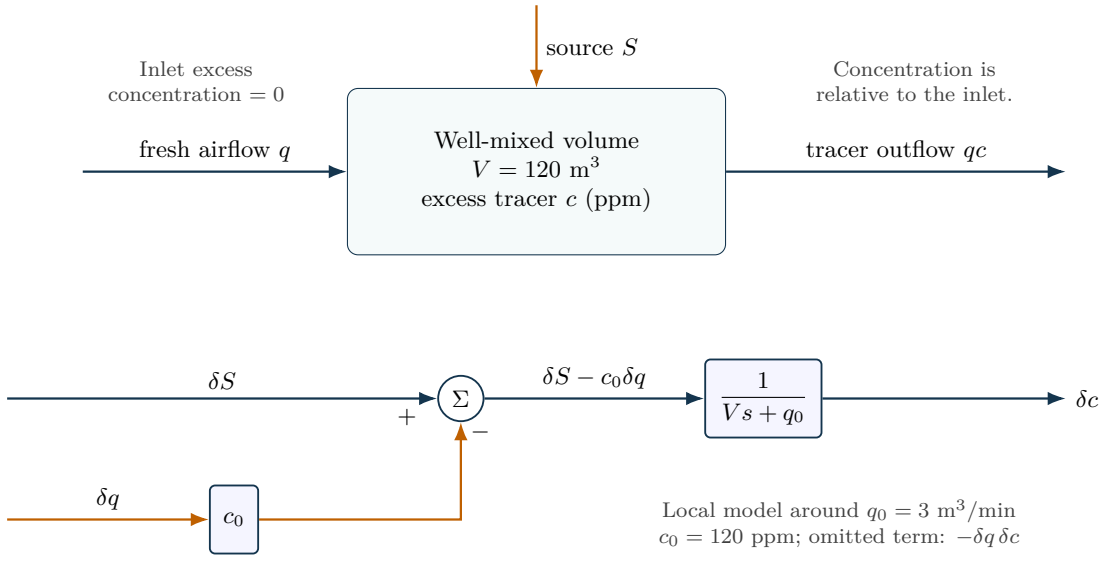
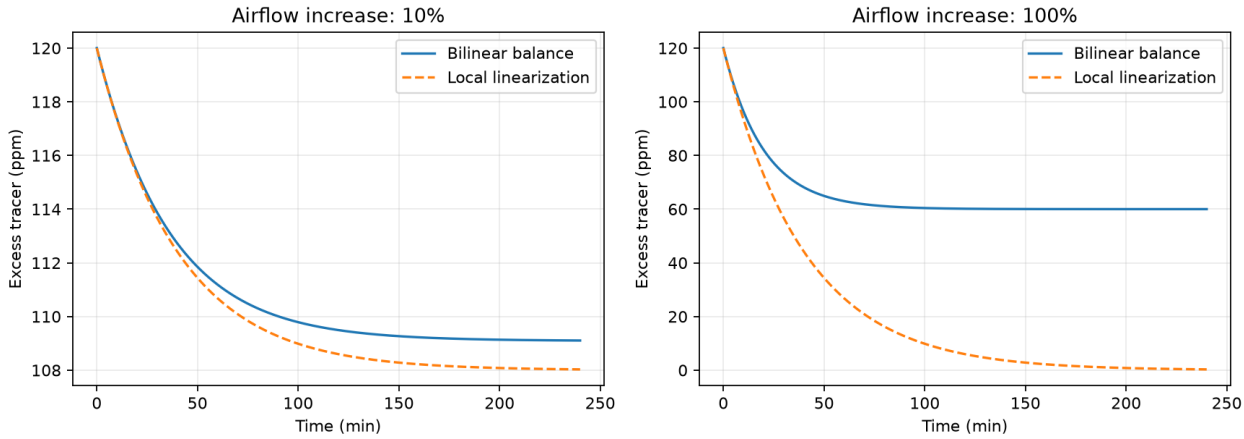


Figure 1. The physical balance and its local two-input signal model. Airflow enters with a negative sign because increased ventilation removes excess tracer.

9 Executed computational evidence

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Airflow +10%: exact final 109.090909 ppm; linear final 108.000000 ppm
Airflow +100%: exact final 60.000000 ppm; linear final 0.000000 ppm
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.